CSE 487 Project Assignment #1
The Impact of Rest on Game-to-Game Performance in Professional Basketball

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1. Problem Statement

1.1 Title

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1.2 Problem Statement

Professional basketball players compete in a demanding schedule that often includes multiple games within a short period of time. As a result, fatigue may influence team performance and outcomes. Many teams implement rest and load management strategies to improve recovery and maintain effectiveness throughout the season. However, the extent to which additional rest affects game-to-game team performance remains unclear.

This project aims to analyze professional basketball data to determine how the number of rest days between games influences team performance. By examining team level statistical indicators such as points scored, assists, rebounds, field goal percentages, turnovers, and game outcomes, the project seeks to identify whether teams perform better when given additional recovery time and whether rest contributes to improved team success.

1.3 Background and Significance

Professional basketball organizations invest significant resources in player health, conditioning, and performance optimization. With the increasing emphasis on load management, understanding the effects of rest has become an important topic for coaches, managers, analysts, and fans. Determining whether additional recovery time leads to measurable improvements in performance can help teams make informed decisions regarding scheduling and game strategy.

1.4 Contribution and Potential Impact

The findings of this project may provide valuable insights for coaches, team managers, sports analysts, and basketball organizations regarding the effectiveness of rest and load management strategies. The results could support decision-making related to player rotations, scheduling considerations, injury prevention efforts, and performance optimization. Additionally, this project demonstrates how data science techniques can be applied to sports analytics to evaluate real-world performance questions using large-scale basketball data.

1.5 Research Questions

1. How does the number of rest days between games affect team performance?

2. Do teams score more points after receiving additional rest?

3. How does rest influence shooting efficiency and turnovers at the team level?

4. Are teams more likely to win after receiving additional recovery time?

5. How do back-to-back games compare to games played after multiple rest days? 6. Can game outcomes be predicted using rest-related variables?

2. Data Sources

The dataset used in this project is the NBA Database from Kaggle, created by Wyatt Walsh. This dataset contains structured professional basketball data, including NBA games, teams, players, and game-level statistics. It is suitable for this project because it provides enough records to analyze game-to-game performance and allows the creation of rest-day variables based on game dates.

The main variables used in this project include game date, team ID, points scored, field goal percentage, turnovers, and game outcome. These variables allow us to compare team performance across different rest-day categories, such as back-to-back games, one day of rest, and multiple days of rest.

The main file used for this phase is game.csv, which contains approximately 65,699 rows before processing. The original file contains both home-team and away-team statistics in each game record. After selecting the relevant columns and converting each game into separate home-team and away-team records, the cleaned team-game dataset contains approximately 131,397 rows and 16 columns.

Dataset source: Walsh, W. NBA Database. Kaggle.
https://www.kaggle.com/datasets/wyattowalsh/basketball

3. Data Cleaning and Processing

The dataset was cleaned and processed using Python. Since this project focuses on the relationship between rest days and basketball performance, the cleaning process focused on preparing team-level game records for chronological analysis. The following six cleaning and processing operations were completed:

1. **Removed duplicate records:** Duplicate game or team-game records were removed to

prevent the same performance from being counted more than once.

2. **Standardized column names:** Column names were converted to a consistent format to make the dataset easier to process in Python.
3. **Converted game dates to datetime format:** Game date columns were converted into datetime objects so that games could be sorted correctly and rest days could be calculated.
4. **Sorted games chronologically by team:** Records were sorted by team and game date to make sure each game was compared with the correct previous game.
5. **Created the rest_days variable:** A new variable, rest_days, was created by calculating the number of days between consecutive games.
6. **Created rest categories:** Rest days were grouped into categories such as 0 days, 1 day, 2 days, and 3+ days. These categories allow easier comparison between back-to-back games and games played after additional recovery time.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the relationship between rest days and basketball performance. The following six EDA operations were conducted:

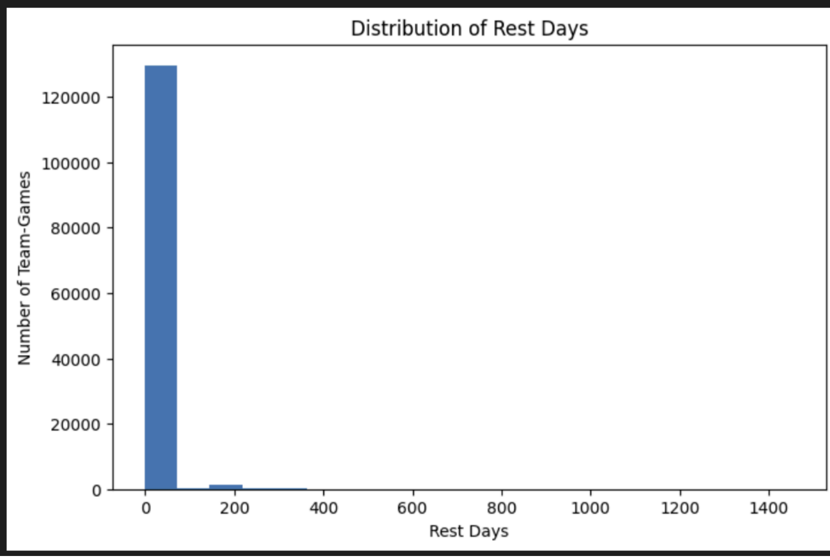
1. **Summary statistics of performance variables:** Summary statistics were calculated for points, rebounds, assists, turnovers, field goal percentage, and rest days.

```python
summary_stats = team_games[["pts", "fg_pct", "reb", "ast", "tov", "rest_days"]].describe()

summary_stats

# EDA 1: Summary statistics
```

	pts	fg_pct	reb	ast	tov	rest_days
count	131396.000000	100417.000000	99942.000000	99790.000000	94027.000000	131314.000000
mean	102.805352	0.461115	42.940505	23.038170	14.991290	3.996367
std	14.701586	0.059645	7.026623	5.612692	4.225784	27.961174
min	18.000000	0.140000	0.000000	0.000000	0.000000	-1.000000
25%	93.000000	0.421000	38.000000	19.000000	12.000000	0.000000
50%	103.000000	0.460000	43.000000	23.000000	15.000000	1.000000
75%	112.000000	0.500000	47.000000	27.000000	18.000000	2.000000
max	196.000000	3.556000	90.000000	89.000000	40.000000	1456.000000

2. **Distribution of rest days:** The distribution of rest days was examined to understand how often teams played back-to-back games compared with games after one or more days of rest.

```
eda_df = team_games[team_games["rest_category"] != "First Game"].copy()

plt.figure(figsize=(8, 5))
plt.hist(eda_df["rest_days"].dropna(), bins=20)
plt.title("Distribution of Rest Days")
plt.xlabel("Rest Days")
plt.ylabel("Number of Team-Games")
plt.show()

# EDA 2: Distribution of rest days
```

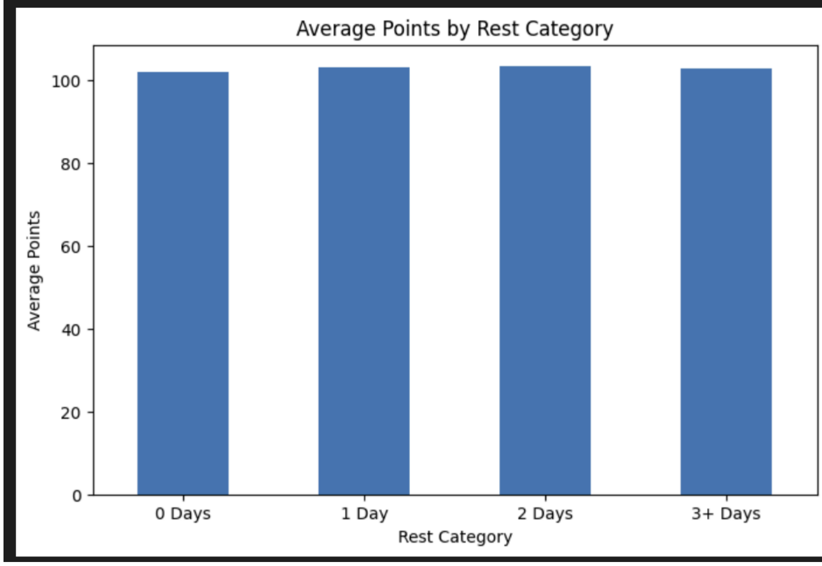


3. **Average points by rest category:** Average points were compared across rest categories to determine whether scoring performance changes with additional rest.

```
points_by_rest = eda_df.groupby("rest_category")["pts"].mean().reindex(
    ["0 Days", "1 Day", "2 Days", "3+ Days"]
)

points_by_rest
# EDA 3: Average points by rest category
```

```
rest_category
0 Days      101.933260
1 Day       103.119006
2 Days      103.282138
3+ Days     102.817847
Name: pts, dtype: float64
```

```python
plt.figure(figsize=(8, 5))
points_by_rest.plot(kind="bar")
plt.title("Average Points by Rest Category")
plt.xlabel("Rest Category")
plt.ylabel("Average Points")
plt.xticks(rotation=0)
plt.show()
# make the bar chart EDA 3
```



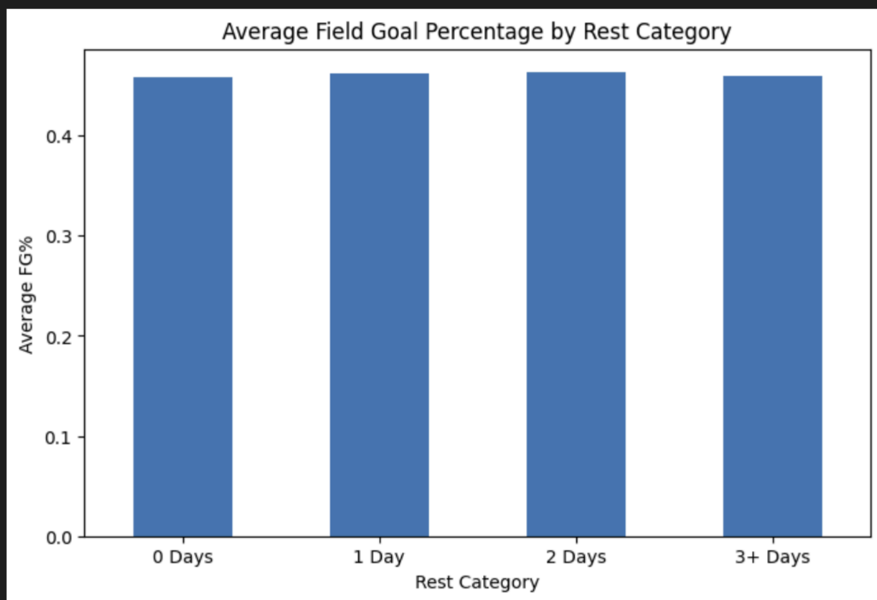
Teams scored slightly fewer points on 0 days of rest compared with games after 1 or 2 days of rest. However, the differences between rest categories are small, suggesting that rest alone may not strongly explain scoring performance.

4. **Shooting percentage by rest category:** Shooting efficiency was analyzed across different rest categories to see whether rest affects field goal percentage.

```python
fg_by_rest = eda_df.groupby("rest_category")["fg_pct"].mean().reindex(
    ["0 Days", "1 Day", "2 Days", "3+ Days"]
)

fg_by_rest
# EDA 4: Shooting percentage by rest category
```

```
rest_category
0 Days      0.457747
1 Day       0.462165
2 Days      0.463202
3+ Days     0.459962
Name: fg_pct, dtype: float64
```

```
plt.figure(figsize=(8, 5))
fg_by_rest.plot(kind="bar")
plt.title("Average Field Goal Percentage by Rest Category")
plt.xlabel("Rest Category")
plt.ylabel("Average FG%")
plt.xticks(rotation=0)
plt.show()
# make the bar chart EDA 4
```

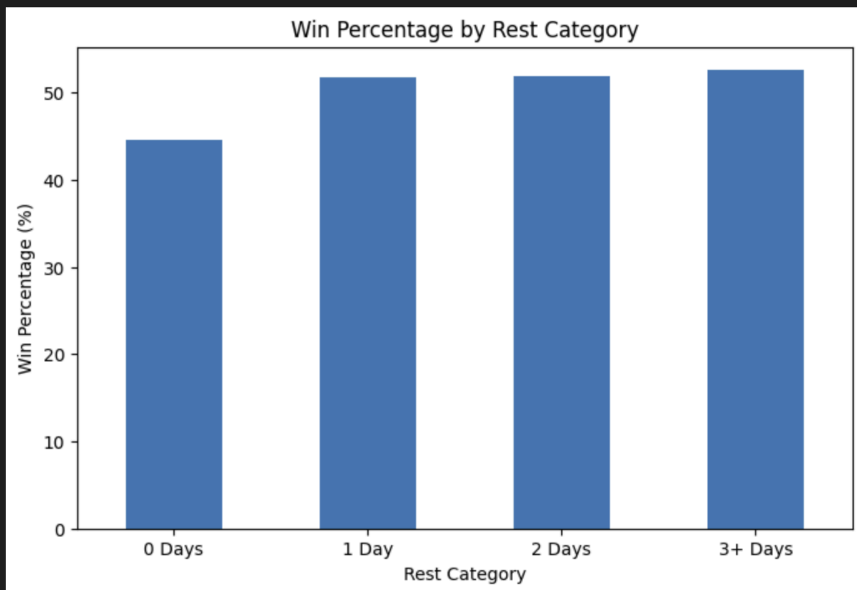


5. **Win percentage by rest category:** Team win percentage was compared across rest categories to examine whether additional rest is associated with better team outcomes.

```
win_by_rest = eda_df.groupby("rest_category")["win"].mean().reindex(
    ["0 Days", "1 Day", "2 Days", "3+ Days"]
) * 100

win_by_rest

# EDA 5: Win percentage by rest category
```

```
rest_category
0 Days     44.548732
1 Day      51.727109
2 Days     51.945776
3+ Days    52.625552
Name: win, dtype: float64
```

```python
plt.figure(figsize=(8, 5))
win_by_rest.plot(kind="bar")
plt.title("Win Percentage by Rest Category")
plt.xlabel("Rest Category")
plt.ylabel("Win Percentage (%)")
plt.xticks(rotation=0)
plt.show()
# make the bar chart EDA 5
```



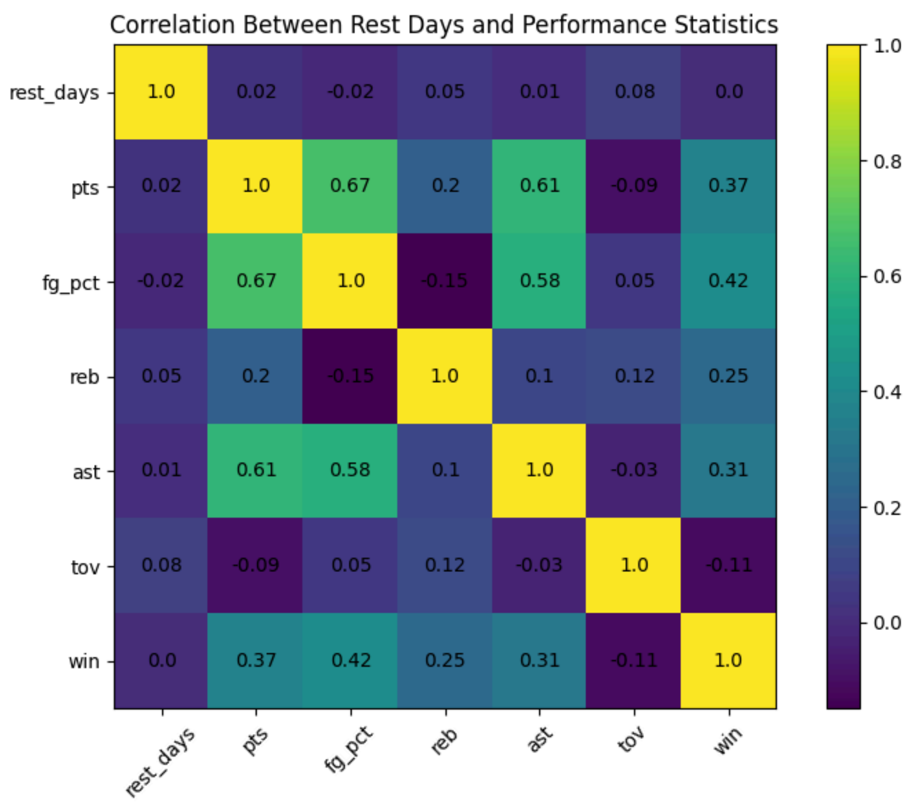
Teams on 0 days of rest had the lowest win percentage, while teams with 1 or more days of rest had higher win percentages. This suggests that rest may have a stronger relationship with game outcome than with raw scoring.

6. **Correlation analysis between rest days and performance statistics:** Correlations were calculated between rest days and team performance variables such as points, turnovers, and shooting percentage.

```python
corr = eda_df[["rest_days", "pts", "fg_pct", "reb", "ast", "tov", "win"]].corr()

corr

# EDA 6: Correlation analysis
```

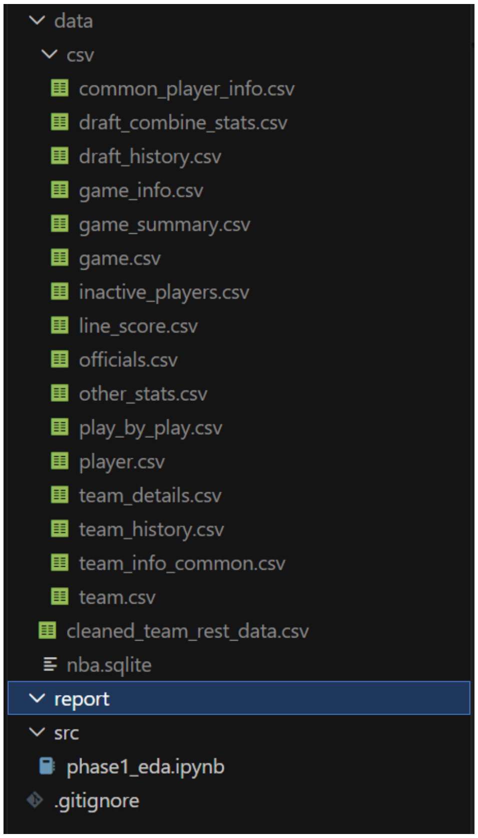
	rest_days	pts	fg_pct	reb	ast	tov	win
rest_days	1.000000	0.017671	-0.016208	0.045573	0.013733	0.076163	0.002767
pts	0.017671	1.000000	0.670003	0.200037	0.613058	-0.092964	0.371830
fg_pct	-0.016208	0.670003	1.000000	-0.149159	0.582583	0.052223	0.423084
reb	0.045573	0.200037	-0.149159	1.000000	0.102913	0.119135	0.248533
ast	0.013733	0.613058	0.582583	0.102913	1.000000	-0.033554	0.308874
tov	0.076163	-0.092964	0.052223	0.119135	-0.033554	1.000000	-0.111361
win	0.002767	0.371830	0.423084	0.248533	0.308874	-0.111361	1.000000

```python
plt.figure(figsize=(8, 6))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Between Rest Days and Performance Statistics")

for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        plt.text(j, i, round(corr.iloc[i, j], 2), ha="center", va="center")

plt.tight_layout()
plt.show()
# turns that table into a heatmap figure
```



The correlation between rest days and most performance variables was weak, meaning rest days alone may not be a strong predictor. However, the rest-category comparison still shows useful differences, especially for win percentage.

Relevant tables and visualizations were created to support these analyses and identify patterns for later modeling. The full Python code for data cleaning, feature engineering, and EDA is included in src/phase1_eda.ipynb. Because the original Kaggle dataset contains very large files, the submitted data folder includes only the necessary cleaned dataset. To reproduce the full workflow from the raw data, users should download the original Kaggle dataset and place the files in the same folder structure shown below.



5. Conclusion

This project seeks to evaluate the impact of rest on game-to-game team performance in professional basketball. Through data cleaning, preprocessing, and exploratory analysis, the project aims to identify meaningful relationships between recovery time and team effectiveness. The findings may provide useful insights for teams, coaches, and analysts regarding load management strategies and performance optimization.

6. References

Walsh, W. NBA Database. Kaggle.
https://www.kaggle.com/datasets/wyattowalsh/basketball